| INCH-POUND |

MIL-M-38510/124C
20 November 1989
SUPERSEDING
MIL-M-38510/124B
5 March 1986

MILITARY SPECIFICATION

MICROCIRCUITS, LINEAR, PRECISION VOLTAGE REFERENCES, MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, precision voltage references. Two product assurance classes and a choice of case outlines and lead finishes are provided and are reflected in the complete Part or Identifying Number (PIN).

1.2 Classification.

1.2.1 Device types. The device types shall be as follows:

Device type	Circuit
01, 04	Precision voltage reference, 6.9 V with temperature stabilizer.
02, 06	Precision voltage reference, 6.9 V without temperature stabilizer.
03	Precision voltage reference, 10 V with guaranteed long term stability.
05	Precision voltage reference, 6.9 V with guaranteed long term stability.
07	Precision voltage reference, 5 V with ultra low drift.
08	Precision voltage reference, 7 V with ultra low drift.
09	Precision voltage reference, 10 V with ultra low drift.

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441-5700, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.2.3 Case outlines. The case outlines shall be designated as follows:

Outline letter	Case outline (see MIL-M-38510, appendix C)
G	A-1 (8-lead, .370" x .185"), can package
X	See figure 1 (2-lead, .219" x .093"), can package
Y	See figure 2 (4-lead, .409" x .234"), can package

1.3 Absolute maximum ratings.

Voltage reference current	
Device types 01, 02, 04, 05, and 06	20 mA
Forward current	
Device types 01, 02, 04, 05, and 06	1 mA
Temperature stabilizer voltage	
Device types 01, 04, and 05	40 V
Reference to substrate voltage	
Device types 01, 04, and 05 (negative terminal of stabilizer)	-0.1 V
Device type 03	
Maximum input voltage	40 V
Minimum input voltage	12 V
Maximum power dissipation	500 mW
Open circuit duration	Indefinite 1/
Device types 07, 08, and 09	
Maximum input voltage	
Device types 07, 08, and 09	44 V
Input-output voltage differential	
Device types 07, 08, and 09	35 V
Output to ground voltage (shunt mode current limit)	
Device types 07 and 08	10 V
Device type 09	16 V
Trim pin to ground voltage	
Positive	Equal to V_O
Negative	-20 V
Device types 07 and 09	
Output short circuit duration	
V_{IN} = 35 V	10 s
$V_{IN} \leq$ 20 V	Indefinite
Device types 07, 08, and 09	
Lead temperature (soldering, 10 seconds)	+300°C
Junction temperature (T_J)	+175°C
Storage temperature range	-55°C to +150°C

1.4 Recommended operating conditions.

Voltage reference current output	
Device type 03	0 mA to 8 mA
Voltage reference current range	
Device types 01 and 02	0.5 mA to 15 mA
Operating ambient temperature range (T_A)	-55°C to +125°C

1.5 Power and thermal characteristics.

Package	Case outline	Maximum allowable power dissipation	Maximum θ_{JC}	Maximum θ_{JC}
8-lead can	G	330 mW at T_A = +125°C	45°C/W	150°C/W
2-lead can	X	115 mW at T_A = +125°C	43°C/W	435°C/W
4-lead can	Y	140 mW at T_A = +125°C	---	200°C/W

1/ Output may be shorted to ground or supply. Rating applies to +125°C case temperature.

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2. APPLICABLE DOCUMENTS

2.1 Government document.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Circuit diagrams and terminal connections. The circuit diagrams and terminal connections shall be as specified on figure 8.

3.2.2 Schematic circuits. The schematic circuits shall be submitted to the preparing activity prior to inclusion of manufacturer's device in this specification and shall be submitted to the qualifying activity and agent activity (DESC-ECS) as a prerequisite for qualification. All qualified manufacturers' schematics shall be maintained by the agent activity and will be available upon request.

3.2.3 Case outlines. The case outlines shall be as specified in 1.2.3.

3.2.4 Package and sealing material. Package and sealing material shall be in accordance with MIL-M-38510.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 (see 6.4).

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I and apply over the full operating ambient temperature range.

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TABLE I. Electrical performance characteristics for device types 01, 02, 04, 05, and 06
(see 3.4 unless otherwise specified.)

Characteristic	Symbol	Conditions		Device type	Limits		Unit
					Min	Max	
Reference voltage	V_R	0.5 mA $\leq I_R \leq$ 10 mA V_S = 30 V	T_A = +25°C	01,04,05	6.80	7.10	V
		0.6 mA $\leq I_R \leq$ 15 mA	T_A = +25°C	02,06	6.70	7.20	
Reference voltage change with current	ΔV_R (current)	0.5 mA $\leq I_R \leq$ 10 mA V_S = 30 V	T_A = +25°C	01,04,05	-9	9	mV
			-55°C $\leq T_A \leq$ +125°C	01,04,05	-12	12	
		0.6 mA $\leq I_R \leq$ 15 mA	T_A = +25°C	02,06	-14	14	
			-55°C $\leq T_A \leq$ +125°C	02,06	-18	18	
Reference voltage temperature	$\frac{\Delta V_R}{\Delta T}$	I_R = 1.0 mA V_S = 30 V	-55°C $\leq T_A \leq$ +85°C	01,05,04	-0.5 -1.0	0.5 1.0	ppm/°C
			+85°C $\leq T_A \leq$ +125°C	01,05,04	-10 -15	10 15	
		I_R = 1.0 mA	-55°C $\leq T_A \leq$ +125°C	02 06	-10 -20	10 20	
Dynamic impedance	Z_D	I_R = 1.0 mA e_i = 1.0 V f = 400 Hz	V_S = 30 V; T_A = +25°C	01,04,05	0	1	Ohm
		(see figure 3)	T_A = +25°C	02,06	0	1	
Noise	N_O	I_R = 1 mA BW = 10 Hz to 10 kHz	V_S = 30 V; T_A = +25°C	01,04,05	0	20	µV RMS
		(see figure 4)	T_A = +25°C	02,06	0	20	
		I_R = 1 mA BW = 0.1 Hz to 10 Hz	V_S = 30 V; T_A = +25°C	01,04,05	0	50	µVp-p
		(see figure 4)	T_A = +25°C	02,06	0	50	

See footnotes at end of table.

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TABLE I. Electrical performance characteristics for device types 01,02,04,05, and 06 - Continued.
(see 3.4 unless otherwise specified)

Characteristic	Symbol	Conditions		Device type	Limits		Unit
					Min	Max	
Temperature stabilizer supply current	I_S	V_S = 40 V I_R = 0 mA	T_A = +25°C	01,04,05	4	20	mA
			T_A = -55°C	01,04,05	6	40	
Initial temperature stabilizer supply current	I_{SI}	9 V ≤ V_S ≤ 40 V I_R = 0 mA	T_A = +25°C	01,04,05	---	200	mA
Reference voltage temperature cycling hysteresis	ΔV_R (temp cycle)	I_R = 1 mA -55°C ≤ T_A ≤ +125°C (see figure 5)	V_S = 30 V	01,04,05	-10	10	mV
				02,06	-1	1	
Long term stability	$\frac{\Delta V_R}{\Delta Time}$	Time = 600 hrs, I_R = 1 mA ±10%	T_A = +25°C V_S = 30 V	05	-20	20	ppm/600 hrs

See footnotes at end of table.

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TABLE I. Electrical performance characteristics for device type 03 - Continued.
(see 3.4 unless otherwise specified.)

Characteristic	Symbol	Conditions (see figure 6, unless otherwise specified)		Device type	Limits		Unit
					Min	Max	
Reference voltage	V_O	V_S = 15 V I_L = 0 mA	T_A = +25°C	03	9.97	10.03	V
			T_A = +125°C	03	9.958	10.042	
			T_A = -55°C	03	9.962	10.038	
Output adjustment range	V_O adj+ V_O adj-	V_S = 15 V I_L = 0 mA	T_A = +25°C	03	0.3 ---	--- -0.3	V
Line regulation	V_{RLINE}	V_S = 13 V to 33 V I_L = 0 mA	T_A = +25°C	03	-0.010	0.010	%/V
			-55°C ≤ T_A ≤ +125°C	03	-0.015	0.015	
Load regulation	V_{RLOAD}	V_S = 15 V I_L = 0 to 10 mA	T_A = +25°C	03	-0.008	0.008	%/mA
		V_S = 15 V I_L = 0 to 8 mA	-55°C ≤ T_A ≤ +125°C	03	-0.012	0.012	
Quiescent supply current	I_{CC}	V_S = 15 V I_L = 0 mA	T_A = +25°C	03	---	1.4	mA
			-55°C ≤ T_A ≤ +125°C	03	---	2.0	
Output short circuit current	I_{OS}	V_S = 15 V V_O = 0 V	T_A = +25°C	03	---	50	mA
Output voltage temperature coefficient	DV_{OUT1}/DT DV_{OUT2}/DT 1/	V_S = 15 V I_L = 0 mA	-55°C ≤ T_A ≤ +25°C	03	-10	10	ppm/°C
			+25°C ≤ T_A ≤ +125°C	03	-12	12	
Output voltage noise	en_{p-p}	V_S = 15 V I_L = 0 mA BW = 0.1 Hz to 10 Hz see figure 7)	T_A = +25°C	03	---	30	µVp-p
Long term drift	V_{LTD} 2/	V_S = 15 V I_L = 0 mA	T_A = +40°C	03	-50	50	ppm/ 1000 hrs

See footnotes at end of table.

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TABLE I. Electrical performance characteristics for device type 07.

Characteristic	Symbol	Conditions: I_{OUT} = 0 mA V_{IN} = 10 V, -55°C ≤ T_A ≤ +125°C, figure 8 (see 3.4, unless otherwise specified)		Limits		Unit
				Min	Max	
Reference voltage	V_O 3/		T_A = +25°C	4.950	5.050	V
			T_A = +125°C	4.9475	5.0525	
			T_A = -55°C	4.948	5.052	
Line regulation	V_{RLINE} 4/	V_{IN} = 10 V, 7.2 V	T_A = +25°C	-0.17	0.17	mV
			T_A = +125°C, -55°C	-0.28	0.28	
		V_{IN} = 10 V, 40 V	T_A = +25°C	-0.9	0.9	
			T_A = +125°C, -55°C	-1.5	1.5	
Load regulation	V_{RLOAD} 4/	I_{OUT} = 0 mA, 10 mA	T_A = +25°C	-5.0	5.0	mV
			T_A = +125°C, -55°C	-7.5	7.5	
		I_{OUT} = 0 mA, -10 mA	T_A = +25°C	-1.0	1.0	
			T_A = +125°C, -55°C	-1.75	1.75	
Trim voltage	V_{TRIM}	I_{TRIM} = 0 mA, T_A = +25°C		-3.5	---	V
Delta voltage/ delta temperature	$\frac{\Delta V_O}{\Delta T}$	T_A = 55°C to +25°C T_A = +25°C to +125°C		--- ---	5.0 5.0	ppm/°C
Noise	N_O 5/	V_{IN} = 15 V, 10 Hz ≤ f ≤ 1 kHz, T_A = +25°C		---	3.5	µV RMS
Supply current	I_{CC}		T_A = +25°C	---	1.2	mA
			T_A = +125°C, -55°C	---	1.5	
Output short circuit current	I_{OS}	V_O = 0 V	T_A = +25°C	-50	---	mA

See footnotes at end of table.

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TABLE I. Electrical performance characteristics for device type 08.

Characteristic	Symbol	Conditions: I_{OUT} = 0 mA, V_{IN} = 12 V, -55°C ≤ T_A ≤ +125°C, figure 8 (see 3.4, unless otherwise specified)		Limits Min	Limits Max	Unit
Reference voltage	V_O 3/		T_A = +25°C	6.950	7.050	V
			T_A = +125°C	6.9465	7.0535	
			T_A = -55°C	6.9472	7.0528	
Line regulation	V_{RLINE} 4/	V_{IN} = 12 V, 8.5 V	T_A = +25°C	-0.1	0.1	mV
			T_A = +125°C, -55°C	-0.2	0.2	
		V_{IN} = 12 V, 40 V	T_A = +25°C	-0.39	0.39	
			T_A = +125°C, -55°C	-0.78	0.78	
Load regulation	V_{RLOAD} 4/	I_{OUT} = 0 mA, -10 mA	T_A = +25°C	-1.75	1.75	mV
			T_A = +125°C, -55°C	-2.8	2.8	
Minimum current shunt mode	I_{MIN}	V_{IN} is open	T_A = +25°C	-1		mA
			T_A = +125°C, -55°C	-1.2		
Shunt mode regulation	V_{SHUNT} 4/ 6/	V_{IN} is open	T_A = +25°C	-6.16	6.16	mV
		I_{OUT} = 1.2 mA, 10 mA	T_A = +125°C, -55°C	-9.24	9.24	
Delta voltage/ delta temperature	$\frac{\Delta V_O}{\Delta T}$	T_A = -55°C to +25°C T_A = +25°C to +125°C		--- ---	5.0 5.0	ppm/°C
Noise	N_O 5/	V_{IN} = 15 V, 10 Hz ≤ f ≤ 1 kHz		---	4	µV RMS
Supply current	I_{CC}		T_A = +25°C	---	1.2	mA
			T_A = +125°C, -55°C	---	1.5	
Output short circuit current	I_{OS}	V_{IN} = 12 V, V_O = 0 V	T_A = +25°C	-50	---	mA

See footnotes at end of table.

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TABLE I. Electrical performance characteristics for device type 09.

Characteristic	Symbol	Conditions: I_{OUT} = 0 mA, V_{IN} = 15 V, -55°C ≤ T_A ≤ +125°C, figure 8 (see 3.4, unless otherwise specified)		Limits		Unit
				Min	Max	
Reference voltage	V_O 3/		T_A = +25°C	9.950	10.050	V
			T_A = +125°C	9.945	10.055	
			T_A = -55°C	9.946	10.054	
Line regulation	V_{RLINE} 4/	V_{IN} = 11.5 V, 14.5 V	T_A = +25°C	-0.12	0.12	mV
			T_A = +125°C, -55°C	-0.18	0.18	
		V_{IN} = 14.5 V, 40 V	T_A = +25°C	-0.5	0.5	
			T_A = +125°C, -55°C	-1.0	1.0	
Load regulation	V_{RLOAD} 4/	I_{OUT} = 0 mA, -10 mA	T_A = +25°C	-2.5	2.5	mV
			T_A = +125°C, -55°C	-4.0	4.0	
Minimum current shunt mode	I_{MIN}	V_{IN} is open	T_A = +25°C	-1.5		mA
			T_A = +125°C, -55°C	-1.7		
Shunt mode regulation	V_{SHUNT} 4/ 6/	I_{OUT} = 1.7 mA, 10 mA	T_A = +25°C	-8.3	8.3	mV
		V_{IN} is open	T_A = +125°C, -55°C	-12.4	12.4	
Trim voltage	V_{TRIM}	I_{TRIM} = 0 mA, T_A = +25°C		-4.3	---	V
Delta voltage/ delta temperature	$\frac{\Delta V_O}{\Delta T}$	T_A = -55°C to +25°C T_A = +25°C to +125°C		--- ---	5.0 5.0	ppm/°C

See footnotes at end of table.

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TABLE I. Electrical performance characteristics for device type 09 - Continued.

Characteristic	Symbol	Conditions: I_{OUT} = 0 mA, V_{IN} = 15 V, -55°C ≤ T_A ≤ +125°C, figure 8 (see 3.4, unless otherwise specified)		Limits		Unit
				Min	Max	
Noise	N_O 5/	V_{IN} = 15 V, 10 Hz ≤ f ≤ 1 kHz, T_A = +25°C		---	6	µV RMS
Supply current	I_{CC}		T_A = +25°C	---	1.7	mA
			T_A = +125°C, -55°C	---	2.0	
Output short circuit current	I_{OS}	V_O = 0 V	T_A = +25°C	-50	---	mA

1/ Output voltage temperature coefficient is defined as the absolute difference between the output voltage at T_A = +125°C and the output voltage at T_A = -55°C divided by the product of the absolute temperature range and the output voltage multiplied by one million, i.e.:

$$DV_{OUT1}/DT = \frac{V_O\ (+25^\circ C) - V_O\ (-55^\circ C)}{(+80^\circ C)\ 10\ V} \times 10^6$$

$$DV_{OUT2}/DT = \frac{V_O\ (+125^\circ C) - V_O\ (+25^\circ C)}{(+100^\circ C)\ 10\ V} \times 10^6$$

2/ Long term drift specification applies only to device under continuous power.

3/ Reference voltage is measured immediately after device is turned ON. Changes due to chip warm up are typically less than 0.005 percent.

4/ Line, load, and shunt mode regulation are measured on a pulse basis. Pulse is not to exceed 30 ms.

5/ RMS noise is measured with a two pole high pass filter at 10 Hz and a two pole low pass filter at 1 kHz.

6/ Shunt voltage is measured with the input open. With the input connected, the shunt mode current can be reduced to 0 mA. Load regulation will remain the same.

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3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.6 Marking. Marking shall be in accordance with MIL-M-38510.

3.6.1 Serialization. All class S devices shall be serialized in accordance with MIL-M-38510.

TABLE II. Electrical test requirements.

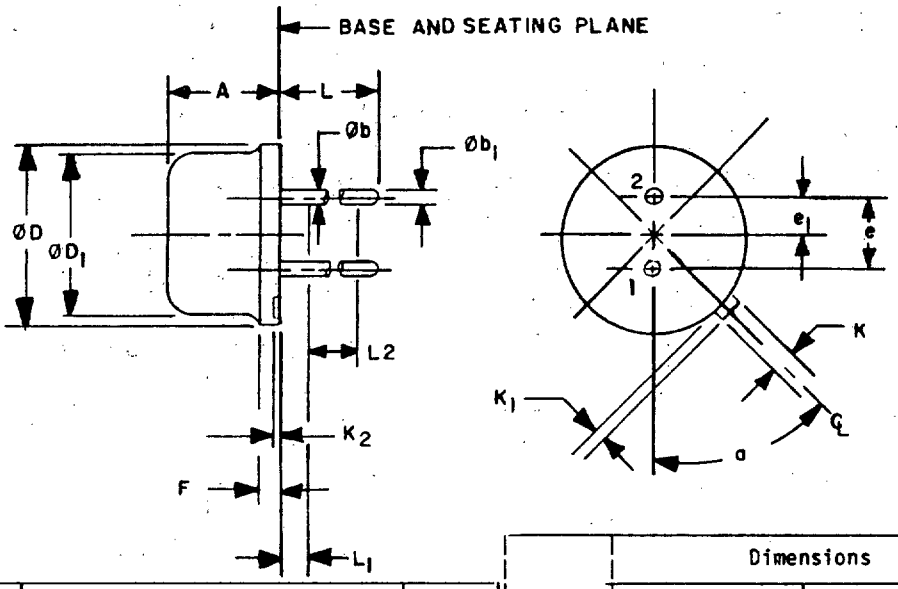
MIL-STD-883 test requirements	Subgroups (see table III)	
	Class S devices	Class B devices 2/
Interim electrical parameters (method 5004)	1	1
Final electrical test parameters 1/ (method 5004)	1,2,3	1,2,3
Group A test requirements (method 5005)	1,2,3,4	1,2,3,4
Group C end-point and group B class S electrical parameters (method 5005)	1,2,3,5, and table IV delta limits	1 and table IV delta limits
Group D end-point electrical parameters (method 5005)	1,2,3	1
Additional electrical subgroups for group D periodic inspections	N/A	5

1/ PDA applies to subgroup 1 (see 4.2c).
2/ Subgroup 4, as defined in table III (tests 18-20, for device types 01,02,04, and 05; tests 13-15, device types 02 and 06) are used for initial and redesign qualification (class B only).

3.6.2 Correctness of indexing and markings. All devices shall be subjected to the final electrical tests specified in table II after part marking to verify that they are correctly indexed and identified by part number. Optionally, an approved electrical test may be devised especially for this requirement.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 59 (see MIL-M-38510, appendix E).

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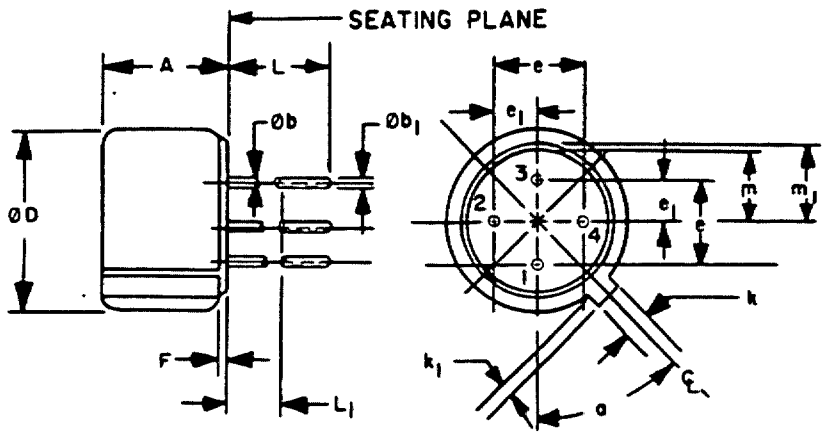
Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
A	.075	.093	1.90	2.36	
øb	.016	.019	0.41	0.48	2
$øb_1$	.016	.021	0.41	0.53	2
øD	.209	.219	5.31	5.56	
$øD_1$	.178	.195	4.52	4.95	
e	.100 BSC		2.54 BSC		4
e_1	.050 BSC		1.27 BSC		4

Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
F	---	.050	---	1.27	
K	.036	.046	0.91	1.17	
K_1	.028	.048	0.71	1.22	3
K_2	.009	.041	0.23	1.04	
L	.500	---	12.70	---	
L_1	---	.050	---	1.27	
L_2	.250	---	6.35	---	
α	45° BSC		45° BSC		4

NOTES:
1. Dimensions are in inches.
2. (All leads) øb applies between L_1 and L_2, $øb_1$ applies between L_2 and .500 (12.70 mm) from the reference plane. Diameter is uncontrolled in L_1 and beyond .500 (12.70 mm) from the reference plane.
3. Measured from the maximum diameter of the product.
4. Leads having a maximum diameter .019 (0.48 mm) measured in gauging plane .054 (1.37 mm) +.001 (0.03 mm) -.000 (0.00 mm) below the base plane of the product shall be within .007 (0.18 mm) of their true-position relative to a maximum width tab.
5. The product may be measured by direct methods or by gauge.
6. Metric equivalents are given for general information only.

FIGURE 1. Case outline X.

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Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
A	.224	.234	5.69	5.94	
øb	.016	.019	0.41	0.48	2
$øb_1$	.016	.021	0.41	0.53	2
øD	.398	.409	10.11	10.39	
e	.100 BSC		2.54 BSC		4
e_1	.050 BSC		1.27 BSC		4

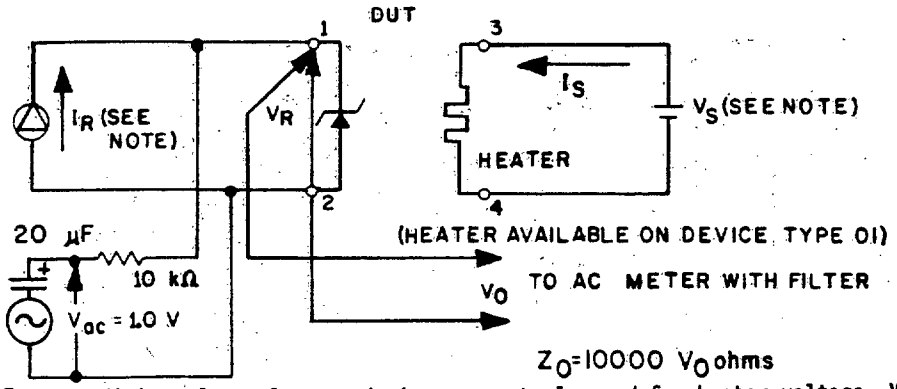
Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
F	.010 BSC		0.25 BSC		
K	.047 BSC		1.19 BSC		
K_1	.008 BSC		0.20 BSC		3
L	.400	---	10.16	---	
L_1	.200	---	5.08	---	
m	.145 BSC		3.68 BSC		
m_1	.175 BSC		4.44 BSC		
α	45° BSC		45° BSC		4

NOTES:
1. Dimensions are in inches.
2. (All leads) øb applies between L_1 and the seating plane. $øb_1$ applies between L_1 and .400 (10.16 mm) from the seating plane.
3. Measured from the maximum diameter of the product.
4. Leads having a maximum diameter .019 (0.48 mm) measured in gauging plane .054 (1.37 mm) +.001 (0.03 mm) -.000 (0.00 mm) below the base plane of the product shall be within .007 (0.18 mm) of their true-position relative to a maximum width tab.
5. The product may be measured by direct methods or by gauge.
6. Metric equivalents are given for general information only.

FIGURE 2. Case outline Y.

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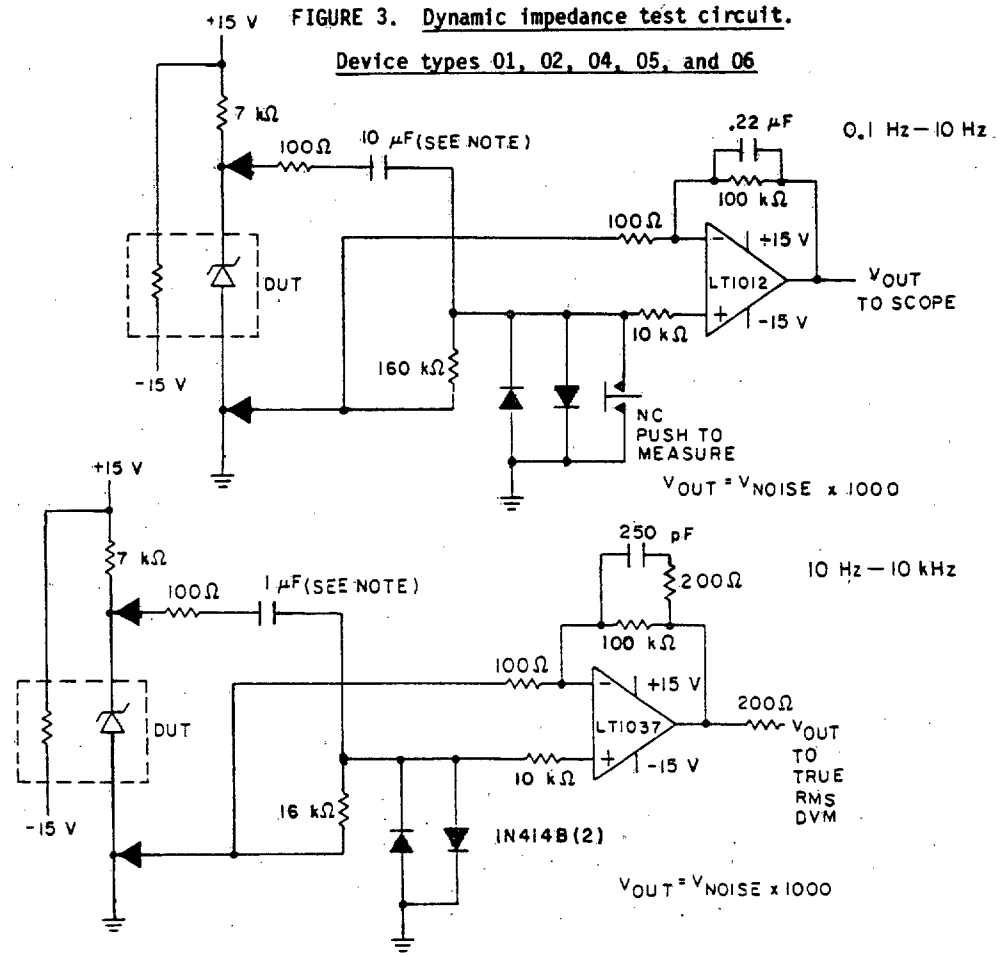
Device types 01, 02, 04, 05, and 06



NOTE: Test conditions for reference device current, I_R, and for heater voltage, V_S are defined in table III.

FIGURE 3. Dynamic impedance test circuit.

Device types 01, 02, 04, 05, and 06

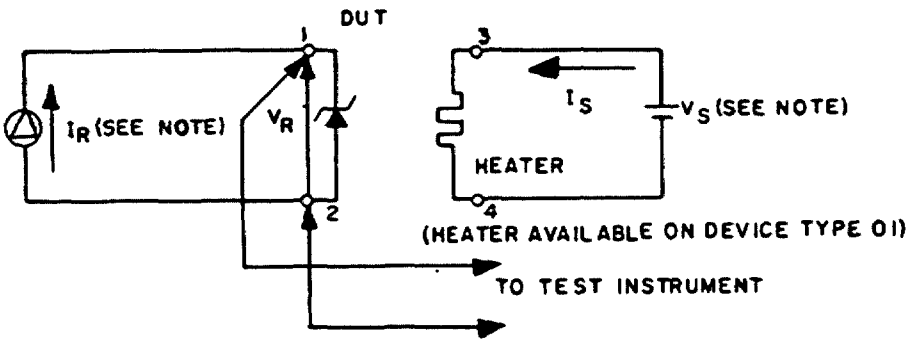


NOTE: Nonelectrolytic.

FIGURE 4. Noise test circuit.

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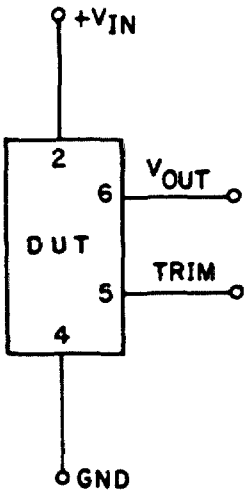
Device types 01, 02, 04, 05, and 06



NOTE: Test conditions for reference device current, I_R, and for heater voltage, V_S are defined in table III.

FIGURE 5. Reference voltage.

Device type 03



Device types 07, 08, and 09

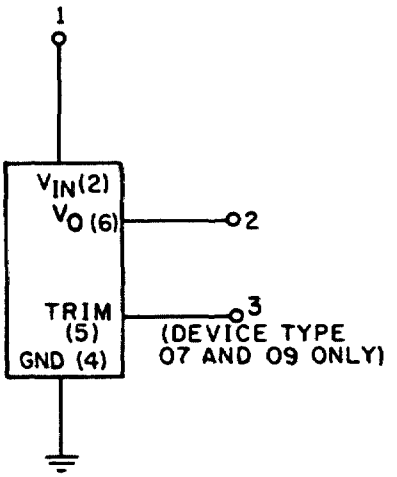
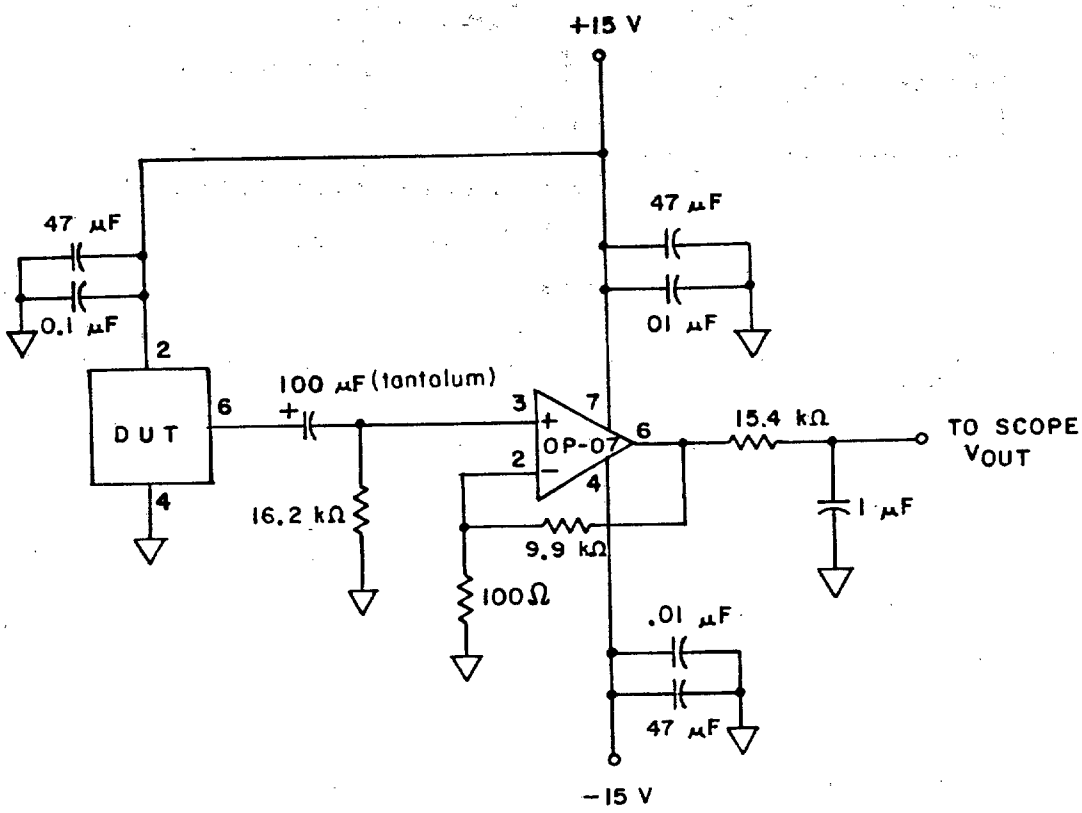


FIGURE 6. Test circuit for static tests.

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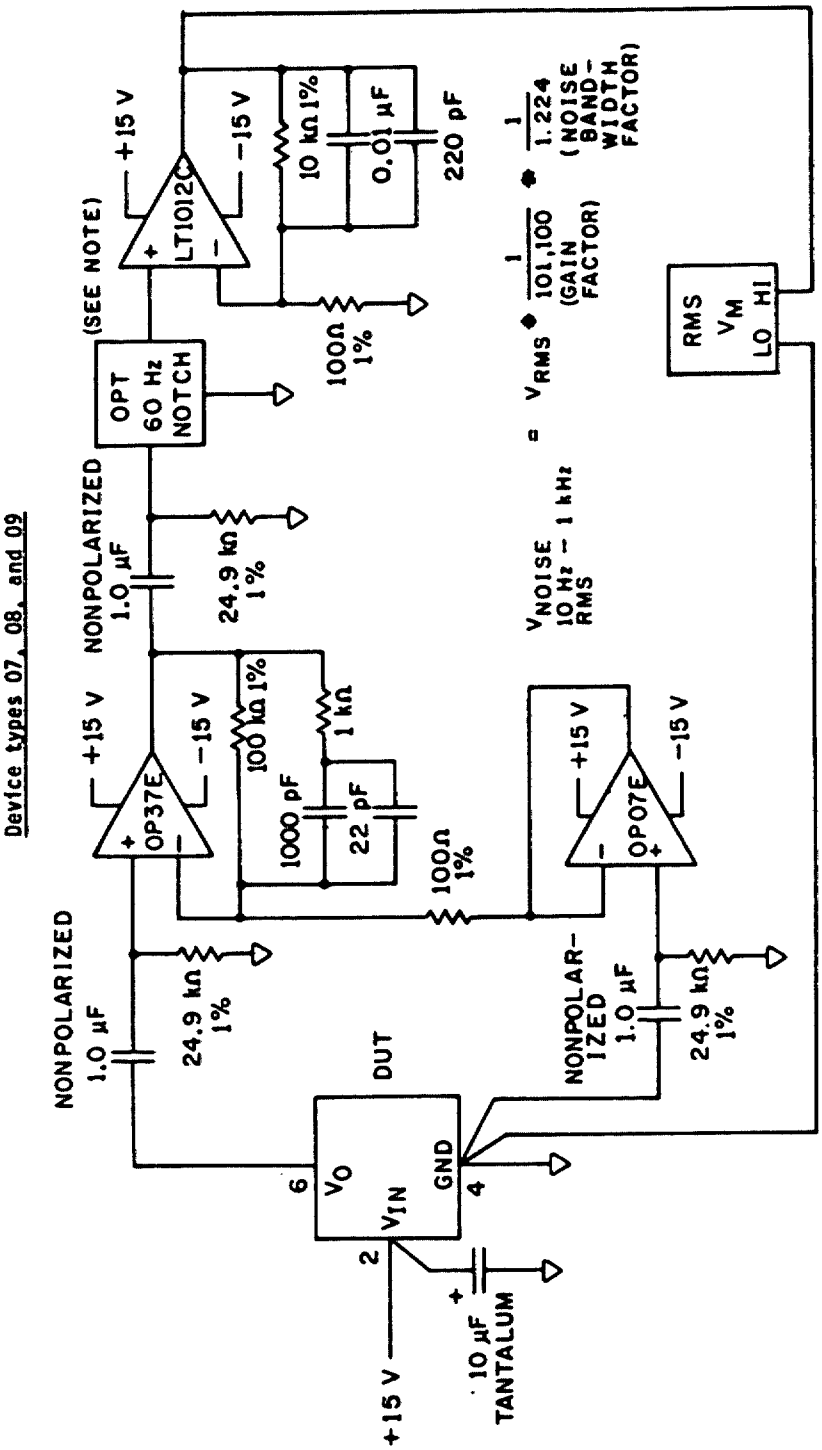
Device type 03



NOTES:
1. Test time = 10 seconds.
2. V_{OUT} measured with differential amplifier 7A22 and lower frequency set to 0.1 Hz.

FIGURE 7. Low frequency noise test circuit.

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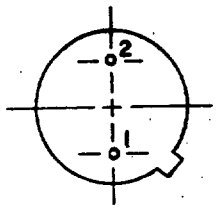
NOTE: The 60 Hz notch filter is used in noisy environments.

FIGURE 7. Low frequency noise test circuit - Continued.

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Device types 02 and 06

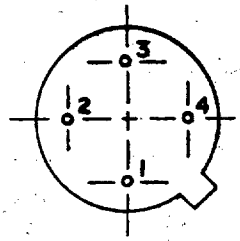
Case X



BOTTOM VIEW

Device types 01, 04, and 05

Case Y



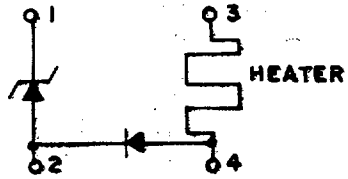
BOTTOM VIEW

Device types 02 and 06



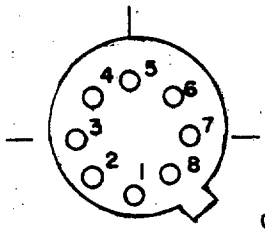
CIRCUIT DIAGRAM

Device types 01, 04, and 05



Device types 03, 07, 08, and 09

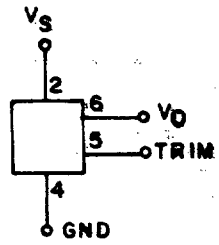
Case G



PIN 4
GROUNDED
TO CASE

BOTTOM VIEW

Device types 03, 07, 08, and 09



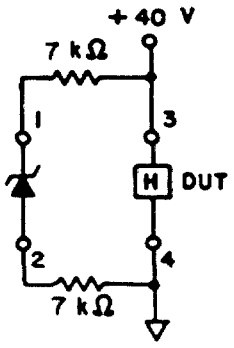
CIRCUIT DIAGRAM

NOTES:
1. No trim pin on device type 08 (do not connect external circuitry to pin 5).
2. For device types 07, 08, and 09, pins 1, 3, 7, and 8 are connected internally. Do not connect external circuitry to these pins.

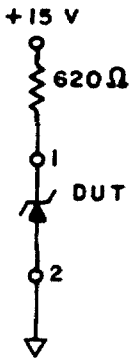
FIGURE 8. Circuit diagrams and terminal connections.

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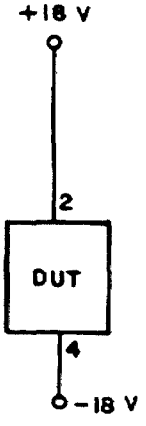
Device types 01, 04, and 05



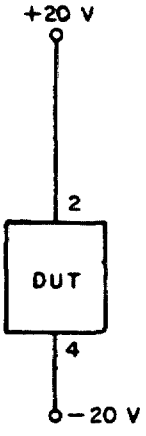
Device types 02 and 06



Device type 03



Device types 07, 08, and 09



NOTE: The actual measured value of the resistor selected shall not exceed ±5 percent of its branded value due to use, heat, or age.

FIGURE 9. Burn-in and steady-state life test circuits.

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TABLE III. Group A inspection for device type 01.

Subgroup	Symbol	Test no.	Conditions (figures 3, 4, and 5, unless otherwise specified) Voltage	Conditions Current	Measured value	Equations	Limits Min	Limits Max	Unit
1 T_A = +25 °C	V_{R1}	1	V_S = 30 V	I_R = .5 mA			6.80	7.10	V
	V_{R2}	2	V_S = 30 V	I_R = 1.0 mA			6.80	7.10	
	V_{R3}	3	V_S = 30 V	I_R = 10 mA			6.80	7.10	
	ΔV_R (current)	4	V_S = 30 V	.5 mA $\leq I_R \leq$ 10 mA		$\Delta V_R = V_{R1}-V_{R3}$	-9	9	mV
	N_O	5	V_S = 30 V BW = 10 Hz to 10 kHz	I_R = 1.0 mA	V_{R4}	$N_O = V_{R4}$	0	20	µV RMS
	I_S	6	V_S = 40 V	I_R = 0 mA			4	20	mA
2 T_A = +85 °C	$\frac{\Delta V_R}{\Delta T}$	7	V_S = 30 V	I_R = 1.0 mA	V_{R5}		---	---	---
2 T_A = +125 °C	ΔV_R (current)	8 9	V_S = 30 V V_S = 30 V	I_R = .5 mA I_R = 10 mA	V_{R6} V_{R7}	$\Delta V_R = V_{R6}-V_{R7}$	-12	12	mV
	$\frac{\Delta V_R}{\Delta T}$	10	V_S = 30 V	I_R = 1.0 mA	V_{R8}	$\frac{\Delta V_R}{\Delta T} = \frac{V_{R8}-V_{R5}}{40}$	-10	10	$\frac{\text{ppm}}{°C}$
3 T_A = -55 °C	ΔV_R (current)	11 12	V_S = 30 V V_S = 30 V	I_R = .5 mA I_R = 10 mA	V_{R9} V_{R10}	$\Delta V_R = V_{R9}-V_{R10}$	-12	12	mV
	$\frac{\Delta V_R}{\Delta T}$	13	V_S = 30 V	I_R = 1.0 mA	V_{R11}	$\frac{\Delta V_R}{\Delta T} = \frac{V_{R11}-V_{R5}}{140}$	-0.5	0.5	$\frac{\text{ppm}}{°C}$
	I_S	14	V_S = 40 V	I_R = 0 mA			6	40	mA
4 T_A = +25 °C	N_O	15	V_S = 30 V BW = 0.1 Hz to 10 Hz	I_R = 1.0 mA	V_{R12}	$N_O = V_{R12}$	0	50	µVp-p
	I_{SI1} I_{SI2}	16 17	V_S = 40 V V_S = 9 V	I_R = 0 mA I_R = 0 mA			---	200	mA
	ΔV_R 1/ (temp cycle)	18 19 20	V_S = 30 V V_S = 30 V V_S = 30 V	I_R = 1 mA I_R = 1 mA I_R = 1 mA	V_{R13} V_{R14} V_{R15}	$\Delta V_R = V_{R13}-V_{R14}$ $\Delta V_R = V_{R14}-V_{R15}$	-10 -10	10 10	mV

1/ Following the measurement (V_{R13}), the temperature T_A shall be cycled to +125°C for 1 hour and returned to +25°C for measurement (V_{R14}). Following the measurement (V_{R14}), the temperature T_A shall be cycled to -55°C for 1 hour and returned to +25°C for measurement (V_{R15}).

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TABLE III. Group A inspection for device type 02.

Subgroup	Symbol	Test no.	Conditions (figures 3, 4, and 5, unless otherwise specified) Voltage	Conditions (figures 3, 4, and 5, unless otherwise specified) Current	Measured value	Equations	Limits Min	Limits Max	Unit
1 T_A = +25°C	V_{R1}	1		I_R = .6 mA			6.70	7.20	V
	V_{R2}	2		I_R = 1.0 mA			6.70	7.20	
	V_{R3}	3		I_R = 15 mA			6.70	7.20	
	ΔV_R (current)	4		.6 mA ≤ I_R ≤ 15 mA		$\Delta V_R = V_{R1}-V_{R3}$	-14	14	mV
	N_O	5	BW = 10 Hz to 10 kHz	I_R = 1.0 mA	V_{R4}	$N_O = V_{R4}$	0	20	µV RMS
2 T_A = +125°C	ΔV_R (current)	6 7		I_R = .6 mA I_R = 15 mA	V_{R5} V_{R6}	$\Delta V_R = V_{R5}-V_{R6}$	-18	18	mV
	$\frac{\Delta V_R}{\Delta T}$	8		I_R = 1.0 mA	V_{R7}	$\frac{\Delta V_R}{\Delta T} = \frac{V_{R7}-V_{R2}}{100}$	-10	10	$\frac{\text{ppm}}{°C}$
3 T_A = -55°C	ΔV_R (current)	9 10		I_R = .6 mA I_R = 15 mA	V_{R8} V_{R9}	$\Delta V_R = V_{R8}-V_{R9}$	-18	18	mV
	$\frac{\Delta V_R}{\Delta T}$	11		I_R = 1.0 mA	V_{R10}	$\frac{\Delta V_R}{\Delta T} = \frac{V_{R2}-V_{R10}}{80}$	-10	10	$\frac{\text{ppm}}{°C}$
4 T_A = +25°C	N_O	12	BW = 0.1 Hz to 10 Hz	I_R = 1.0 mA	V_{R11}	$N_O = V_{R11}$	0	50	µVp-p
	ΔV_R 1/ (temp cycle)	13 14 15		I_R = 1.0 mA I_R = 1.0 mA I_R = 1.0 mA	V_{R12} V_{R13} V_{R14}	$\Delta V_R = V_{R12}-V_{R13}$ $\Delta V_R = V_{R13}-V_{R14}$	-1 -1	1 1	mV

1/ Following the measurement (V_{R12}), the temperature T_A shall be cycled to +125°C for 1 hour and returned to +25°C for measurement (V_{R13}). Following the measurement (V_{R13}), the temperature T_A shall be cycled to -55°C for 1 hour and returned to +25°C for measurement (V_{R14}).

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TABLE III. Group A inspection for device type 03.

Subgroup	Symbol	Test no.	Conditions (figures 6 and 7, unless otherwise specified) Voltage	Conditions (figures 6 and 7, unless otherwise specified) Current	Measured value	Equations	Limits Min	Limits Max	Units
1 T_A = +25°C	I_{CC}	1	V_S = 15 V	I_L = 0 mA			---	1.4	mA
	V_O	2	V_S = 15 V	I_L = 0 mA	V_{01}		9.970	10.030	V
	V_O adj+	3	TRIM 10 V V_S = 15 V	I_L = 0 mA	V_{02}	$(V_{01} - V_{02})$ 10	0.3	---	V
	V_O adj-	4	TRIM 10 V V_S = 15 V	I_L = 0 mA	V_{03}	$(V_{01} - V_{03})$ 10	---	-0.3	V
	V_{RLINE}	5	V_S = 13 V	I_L = 0 mA	V_{04}	$V_{LINE} = \frac{V_{04} - V_{05}}{(10\ V)(20\ V)}$ (100)	-0.010	0.010	%/V
		6	V_S = 33 V	I_L = 0 mA	V_{05}				
	V_{RLOAD}	7	V_S = 15 V	I_L = 0 mA	V_{06}	$V_{RLOAD} = \frac{V_{06} - V_{07}}{(10\ V)(10\ mA)}$ (100)	-0.008	0.008	%/mA
		8	V_S = 15 V	I_L = 10 mA	V_{07}				
	I_{OS}	9	V_S = 15 V V_O = 0 V				---	50	mA
2 T_A = +125°C	I_{CC}	10	V_S = 15 V	I_L = 0 mA			---	2.0	mA
	V_O	11	V_S = 15 V	I_L = 0 mA	V_{08}		9.958	10.042	V
	V_{RLINE}	12	V_S = 15 V	I_L = 0 mA	V_{09}	$V_{LINE} = \frac{V_{09} - V_{010}}{(10\ V)(20\ V)}$ (100)	-0.015	0.015	%/V
		13	V_S = 33 V	I_L = 0 mA	V_{010}				
	V_{RLOAD}	14	V_S = 15 V	I_L = 0 mA	V_{011}	$V_{LOAD} = \frac{V_{011} - V_{012}}{(10\ V)\ (8\ mA)}$ (100)	-0.012	0.012	%/mA
		15	V_S = 15 V	I_L = 8 mA	V_{012}				
	DV_{OUT}/DT	16	V_S = 15 V	I_L = 0 mA	V_{013}	$DV_{OUT}/DT = \frac{V_{013} - V_{01}}{(100°C)\ (10\ V)}$ X10^6	-12	12	ppm/°C

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TABLE III. Group A inspection for device type 03 - Continued.

Subgroup	Symbol	Test no.	Conditions (figures 6 and 7, unless otherwise specified) Voltage	Conditions (figures 6 and 7, unless otherwise specified) Current	Measured value	Equations	Limits Min	Limits Max	Units
3 T_A = -55°C	I_{CC}	17	V_S = 15 V	I_L = 0 mA			---	2.0	mA
	V_O	18	V_S = 15 V	I_L = 0 mA	V_{014}		9.962	10.038	V
	V_{RLINE}	19	V_S = 13 V	I_L = 0 mA	V_{015}	$V_{RLINE} = \frac{V_{015} - V_{016}}{(10\ V)\ (20\ V)}\ (100)$	-0.015	0.015	%/V
		20	V_S = 33 V	I_L = 0 mA	V_{016}				
	V_{RLOAD}	21	V_S = 15 V	I_L = 0 mA	V_{017}	$V_{RLOAD} = \frac{V_{017} - V_{018}}{(10\ V)\ (8\ mA)}\ (100)$	-0.012	0.012	%/mA
		22	V_S = 15 V	I_L = 8 mA	V_{018}				
	DV_{OUT}/DT	23	V_S = 15 V	I_L = 0 mA	V_{019}	$DV_{OUT}/DT = \frac{V_{01} - V_{019}}{(80°C)\ (10\ V)} \times 10^6$	-10	10	ppm/°C
4 T_A = +25°C	e_{np-p}	24	V_S = 15 V	I_L = 0 mA			---	30	µVp-p
			B_W = 0.1 Hz to 10 Hz						
5 T_A = +40°C	V_{LTD}	25	V_S = 15 V	I_L = 0 mA		Device under continuous power for 1000 hours.	-50	50	ppm/1000 hrs

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TABLE III. Group A inspection for device type 04.

Subgroup	Symbol	Test no.	Conditions (figures 3, 4, and 5, unless otherwise specified) Voltage	Conditions Current	Measured value	Equations	Limits Min	Limits Max	Unit
1 T_A = +25°C	V_{R1}	1	V_S = 30 V	I_R = .5 mA			6.80	7.10	V
	V_{R2}	2	V_S = 30 V	I_R = 1.0 mA			6.80	7.10	
	V_{R3}	3	V_S = 30 V	I_R = 10 mA			6.80	7.10	
	ΔV_R (current)	4	V_S = 30 V	.5 mA $\leq I_R \leq$ 10 mA		$\Delta V_R = V_{R1}-V_{R3}$	-9	9	mV
	N_O	5	V_S = 30 V, BW = 10 Hz to 10 kHz	I_R = 1.0 mA	V_{R4}	$N_O = V_{R4}$	0	20	µV RMS
	I_S	6	V_S = 40 V	I_R = 0 mA			4	20	mA
2 T_A = +85°C	$\frac{\Delta V_R}{\Delta T}$	7	V_S = 30 V	I_R = 1.0 mA	V_{R5}		---	---	---
2 T_A = +125°C	ΔV_R (current)	8 9	V_S = 30 V V_S = 30 V	I_R = .5 mA I_R = 10 mA	V_{R6} V_{R7}	$\Delta V_R = V_{R6}-V_{R7}$	-12	12	mV
	$\frac{\Delta V_R}{\Delta T}$	10	V_S = 30 V	I_R = 1.0 mA	V_{R8}	$\frac{\Delta V_R}{\Delta T} = \frac{V_{R8}-V_{R5}}{40}$	-15	15	$\frac{\text{ppm}}{°C}$
3 T_A = -55°C	ΔV_R (current)	11 12	V_S = 30 V V_S = 30 V	I_R = .5 mA I_R = 10 mA	V_{R9} V_{R10}	$\Delta V_R = V_{R9}-V_{R10}$	-12	12	mV
	$\frac{\Delta V_R}{\Delta T}$	13	V_S = 30 V	I_R = 1.0 mA	V_{R11}	$\frac{\Delta V_R}{\Delta T} = \frac{V_{R11}-V_{R5}}{140}$	-1.0	1.0	$\frac{\text{ppm}}{°C}$
	I_S	14	V_S = 40 V	I_R = 0 mA			6	40	mA
4 T_A = +25°C	N_O	15	V_S = 30 V, BW = 0.1 Hz to 10 Hz	I_R = 1.0 mA	V_{R12}	$N_O = V_{R12}$	0	50	µVp-p
	I_{SI1} I_{SI2}	16 17	V_S = 40 V V_S = 9 V	I_R = 0 mA I_R = 0 mA			---	200	mA
	ΔV_R 1/ (temp cycle)	18 19 20	V_S = 30 V V_S = 30 V V_S = 30 V	I_R = 1 mA I_R = 1 mA I_R = 1 mA	V_{R13} V_{R14} V_{R15}	$\Delta V_R = V_{R13}-V_{R14}$ $\Delta V_R = V_{R14}-V_{R15}$	-10 -10	10 10	mV

1/ Following the measurement (V_{R13}), the temperature T_A shall be cycled to +125°C for 1 hour and returned to +25°C for measurement (V_{R14}). Following the measurement (V_{R14}), the temperature T_A shall be cycled to -55°C for 1 hour and returned to +25°C for measurement (V_{R15}).

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TABLE III. Group A inspection for device type 05.

Subgroup	Symbol	Test no.	Conditions (figures 3, 4, and 5, unless otherwise specified) Voltage	Conditions (figures 3, 4, and 5, unless otherwise specified) Current	Measured value	Equations	Limits Min	Limits Max	Unit
1 T_A = +25°C	V_{R1}	1	V_S = 30 V	I_R = .5 mA			6.80	7.10	V
	V_{R2}	2	V_S = 30 V	I_R = 1.0 mA			6.80	7.10	
	V_{R3}	3	V_S = 30 V	I_R = 10 mA			6.80	7.10	
	ΔV_R (current)	4	V_S = 30 V	.5 mA $\leq I_R \leq$ 10 mA		$\Delta V_R = V_{R1}-V_{R3}$	-9	9	mV
	N_O	5	V_S = 30 V BW = 10 Hz to 10 kHz	I_R = 1.0 mA	V_{R4}	$N_O = V_{R4}$	0	20	µV RMS
	I_S	6	V_S = 40 V	I_R = 0 mA			4	20	mA
2 T_A = +85°C	$\frac{\Delta V_R}{\Delta T}$	7	V_S = 30 V	I_R = 1.0 mA	V_{R5}		---	---	---
2 T_A = +125°C	ΔV_R (current)	8 9	V_S = 30 V V_S = 30 V	I_R = .5 mA I_R = 10 mA	V_{R6} V_{R7}	$\Delta V_R = V_{R6}-V_{R7}$	-12	12	mV
	$\frac{\Delta V_R}{\Delta T}$	10	V_S = 30 V	I_R = 1.0 mA	V_{R8}	$\frac{\Delta V_R}{\Delta T} = \frac{V_{R8}-V_{R5}}{40}$	-10	10	$\frac{\text{ppm}}{°C}$
3 T_A = -55°C	ΔV_R (current)	11 12	V_S = 30 V V_S = 30 V	I_R = .5 mA I_R = 10 mA	V_{R9} V_{R10}	$\Delta V_R = V_{R9}-V_{R10}$	-12	12	mV
	$\frac{\Delta V_R}{\Delta T}$	13	V_S = 30 V	I_R = 1.0 mA	V_{R11}	$\frac{\Delta V_R}{\Delta T} = \frac{V_{R11}-V_{R5}}{140}$	-0.5	0.5	$\frac{\text{ppm}}{°C}$
	I_S	14	V_S = 40 V	I_R = 0 mA			6	40	mA
4 T_A = +25°C	N_O	15	V_S = 30 V BW = 0.1 Hz to 10 Hz	I_R = 1.0 mA	V_{R12}	$N_O = V_{R12}$	0	50	µVp-p
	I_{S11} I_{S12}	16 17	V_S = 40 V V_S = 9 V	I_R = 0 mA I_R = 0 mA			---	200	mA
	ΔV_R 1/ (temp cycle)	18 19 20	V_S = 30 V V_S = 30 V V_S = 30 V	I_R = 1 mA I_R = 1 mA I_R = 1 mA	V_{R13} V_{R14} V_{R15}	$\Delta V_R = V_{R13}-V_{R14}$ $\Delta V_R = V_{R14}-V_{R15}$	-10 -10	10 10	mV
5 T_A = +25°C	$\frac{\Delta V_R}{\Delta \text{time}}$	21	V_S = 30 V	I_R = 1 mA	V_{R16}	$\Delta V_R = V_{R16}-V_{R2}$	-20	20	$\frac{\text{ppm}}{\text{600 hours}}$

1/ Following the measurement (V_{R13}), the temperature T_A shall be cycled to +125°C for 1 hour and returned to +25°C for measurement (V_{R14}). Following the measurement (V_{R14}), the temperature T_A shall be cycled to -55°C for 1 hour and returned to +25°C for measurement (V_{R15}).

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TABLE III. Group A inspection for device type 06.

Subgroup	Symbol	Test no.	Conditions (figures 3, 4, and 5, unless otherwise specified) Voltage	Conditions (figures 3, 4, and 5, unless otherwise specified) Current	Measured value	Equations	Limits Min	Limits Max	Unit
1 T_A = +25°C	V_{R1}	1		I_R = .6 mA			6.70	7.20	V
	V_{R2}	2		I_R = 1.0 mA			6.70	7.20	
	V_{R3}	3		I_R = 15 mA			6.70	7.20	
	ΔV_R (current)	4		.6 mA $\leq I_R \leq$ 15 mA		$\Delta V_R = V_{R1}-V_{R3}$	-14	14	mV
	N_O	5	BW = 10 Hz to 10 kHz	I_R = 1.0 mA	V_{R4}	$N_O = V_{R4}$	0	20	µV RMS
2 T_A = +125°C	ΔV_R (current)	6 7		I_R = .6 mA I_R = 15 mA	V_{R5} V_{R6}	$\Delta V_R = V_{R5}-V_{R6}$	-18	18	mV
	$\frac{\Delta V_R}{\Delta T}$	8		I_R = 1.0 mA	V_{R7}	$\frac{\Delta V_R}{\Delta T} = \frac{V_{R7}-V_{R2}}{100}$	-20	20	$\frac{\text{ppm}}{°C}$
3 T_A = -55°C	ΔV_R (current)	9 10		I_R = .6 mA I_R = 15 mA	V_{R8} V_{R9}	$\Delta V_R = V_{R8}-V_{R9}$	-18	18	mV
	$\frac{\Delta V_R}{\Delta T}$	11		I_R = 1.0 mA	V_{R10}	$\frac{\Delta V_R}{\Delta T} = \frac{V_{R2}-V_{R10}}{80}$	-20	20	$\frac{\text{ppm}}{°C}$
4 T_A = +25°C	N_O	12	BW = 0.1 Hz to 10 Hz	I_R = 1.0 mA	V_{R11}	$N_O = V_{R11}$	0	50	µVp-p
	ΔV_R 1/ (temp cycle)	13 14 15		I_R = 1.0 mA I_R = 1.0 mA I_R = 1.0 mA	V_{R12} V_{R13} V_{R14}	$\Delta V_R = V_{R12}-V_{R13}$ $\Delta V_R = V_{R13}-V_{R14}$	-1 -1	1 1	mV

1/ Following the measurement (V_{R12}), the temperature T_A shall be cycled to +125°C for 1 hour and returned to +25°C for measurement (V_{R13}). Following the measurement (V_{R13}), the temperature T_A shall be cycled to -55°C for 1 hour and returned to +25°C for measurement (V_{R14}).

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TABLE III. Group A inspection for device type 07.

Subgroups	Symbol	Test no.	Measurement value	Adapter pin number See figure 6			Pin measurement	Equations	Limits		Unit
				1	2	3			Min	Max	
1	I_{CC}	1	I_1	10 V	0 mA	0 mA	1	$I_{CC} = I_1$	---	1.2	mA
$T_A = +25°C$	V_O 1/	2	V_1	10 V	0 mA	0 mA	2	$V_O = V_1$	4.950	5.050	V
	V_{RLINE} 2/	3	V_2	7.2 V	0 mA	0 mA	2	$V_{RLINE} = V_1 - V_2$	-0.17	0.17	mV
		4	V_3	40 V	0 mA	0 mA	2	$V_{RLINE} = V_1 - V_3$	-0.9	0.9	mV
	V_{RLOAD} 2/	5	V_4	10 V	0 mA	0 mA	2				
		6	V_5	10 V	10 mA	0 mA	2	$V_{RLOAD} = V_4 - V_5$	-5.0	5.0	mV
		7	V_6	10 V	-10 mA	0 mA	2	$V_{RLOAD} = V_4 - V_6$	-1.0	1.0	mV
	V_{TRIM}	8	V_7	10 V	0 mA	0 mA	3	$V_{TRIM} = V_7$	3.5	---	V
	I_{OS}	9	I_2	10 V	0 mA	0 mA	2	$I_{OS} = I_2$	-50	---	mA
2	I_{CC}	10	I_3	10 V	0 mA	0 mA	1	$I_{CC} = I_3$	---	1.5	mA
$T_A = +125°C$	V_O 1/	11	V_8	10 V	0 mA	0 mA	2	$V_O = V_8$	4.9475	5.0525	V
	V_{RLINE} 2/	12	V_9	7.2 V	0 mA	0 mA	2	$V_{RLINE} = V_8 - V_9$	-0.28	0.28	mV
		13	V_{10}	40 V	0 mA	0 mA	2	$V_{RLINE} = V_8 - V_{10}$	-1.5	1.5	mV
	V_{RLOAD} 2/	14	V_{11}	10 V	0 mA	0 mA	2				
		15	V_{12}	10 V	10 mA	0 mA	2	$V_{RLOAD} = V_{11} - V_{12}$	-7.5	7.5	mV
		16	V_{13}	10 V	-10 mA	0 mA	2	$V_{RLOAD} = V_{11} - V_{13}$	-1.75	1.75	mV
	$\frac{\Delta V_O}{\Delta T}$	17						$\frac{\Delta V_O}{\Delta T} = \frac{V_8 - V_1}{100}$	---	5	ppm/°C

See footnotes at end of table for device type 07.

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TABLE III. Group A inspection for device type 07 - Continued.

Subgroups	Symbol	Test no.	Measurement value	Adapter pin number See figure 6			Pin measurement	Equations	Limits		Unit
				1	2	3			Min	Max	
3 T_A = -55°C	I_{CC}	18	I_4	10 V	0 mA	0 mA	1	$I_{CC} = I_4$	---	1.5	mA
	V_O 1/	19	V_{14}	10 V	0 mA	0 mA	2	$V_O = V_{14}$	4.948	5.052	V
	V_{RLINE} 2/	20 21	V_{15} V_{16}	7.2 V 40 V	0 mA 0 mA	0 mA 0 mA	2 2	$V_{RLINE} = V_{14} - V_{15}$ $V_{RLINE} = V_{14} - V_{16}$	-0.28 -1.5	0.28 1.5	mV mV
	V_{RLOAD} 2/	22 23 24	V_{17} V_{18} V_{19}	10 V 10 V 10 V	0 mA 10 mA -10 mA	0 mA 0 mA 0 mA	2 2 2	$V_{RLOAD} = V_{17} - V_{18}$ $V_{RLOAD} = V_{17} - V_{19}$	-7.5 -1.75	7.5 1.75	mV mV
	$\frac{\Delta V_O}{\Delta T}$	25						$\frac{\Delta V_O}{\Delta T} = \frac{V_{14} - V_1}{80}$	---	5	ppm/°C
4 T_A = +25°C	N_O 3/	26	N_1	15 V	0 mA	0 mA	---	See figure 7	---	3.5	µV RMS

1/ Reference voltage is measured immediately after device is turned ON. Changes due to chip warm up are typically less than 0.005 percent.

2/ Line and load regulation are measured on a pulse basis. Pulse is not to exceed 30 ms.

3/ RMS noise is measured with a two pole high pass filter at 10 Hz and a two pole low pass filter at 1 kHz.

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TABLE III. Group A inspection for device type 08.

Subgroups	Symbol	Test no.	Measurement value	Adapter pin number See figure 6		Pin measurement	Equations	Limits		Unit
				1	2			Min	Max	
1	I_{CC}	1	I_1	12 V	0 mA	1	$I_{CC} = I_1$	---	1.2	mA
T_A = +25°C	V_O 1/	2	V_1	12 V	0 mA	2	$V_O = V_1$	6.950	7.050	V
	V_{RLINE} 2/	3	V_2	8.5 V	0 mA	2	$V_{RLINE} = V_1 - V_2$	-0.1	0.1	mV
		4	V_3	40 V	0 mA	2	$V_{RLINE} = V_1 - V_3$	-0.39	0.39	mV
	V_{RLOAD} 2/	5	V_4	12 V	0 mA	2				
		6	V_5	12 V	-10 mA	2	$V_{RLOAD} = V_4 - V_5$	-1.75	1.75	mV
	I_{MIN}	7	V_6	0 mA	2 mA	2	$V_7 = V_6 - 50$ mV	---	---	mA
		8	I_2	0 mA	V_7	2	$I_{MIN} = I_2$	-1	---	
	V_{SHUNT} 2/ 3/	9	V_8	0 mA	1.2 mA	2	$V_{SHUNT} = V_9 - V_8$	-6.16	6.16	mV
		10	V_9	0 mA	10 mA	2				
	I_{OS}	11	I_3	12 V	0 V	2	$I_{OS} = I_3$	-50	---	mA
2	I_{CC}	12	I_4	12 V	0 mA	1	$I_{CC} = I_4$	---	1.5	mA
T_A = +125°C	V_O 1/	13	V_{10}	12 V	0 mA	2	$V_O = V_{10}$	6.9465	7.0535	V
	V_{RLINE} 2/	14	V_{11}	8.5 V	0 mA	2	$V_{RLINE} = V_{10} - V_{11}$	-0.2	0.2	mV
		15	V_{12}	40 V	0 mA	2	$V_{RLINE} = V_{10} - V_{12}$	-0.78	0.78	mV
	V_{RLOAD} 2/	16	V_{13}	12 V	0 mA	2				
		17	V_{14}	12 V	-10 mA	2	$V_{RLOAD} = V_{13} - V_{14}$	-2.8	2.8	mV
	I_{MIN}	18	V_{15}	0 mA	2 mA	2	$V_{16} = V_{15} - 50$ mV	---	---	mA
		19	I_5	0 mA	V_{16}	2	$I_{MIN} = I_5$	-1.2	---	

See footnotes at end of table for device type 08.

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TABLE III. Group A inspection for device type 08 - Continued.

Subgroups	Symbol	Test no.	Measurement value	Adapter pin number See figure 6		Pin measurement	Equations	Limits		Unit
				1	2			Min	Max	
2 T_A = +125°C	V_{SHUNT} 2/ 3/	20 21	V_{17} V_{18}	0 mA 0 mA	1.2 mA 10 mA	2 2	$V_{SHUNT} = V_{18} - V_{17}$	-9.24	9.24	mV
	$\frac{\Delta V_O}{\Delta T}$	22					$\frac{\Delta V_O}{\Delta T} = \frac{V_{10} - V_1}{100}$	---	5	ppm/°C
3 T_A = -55°C	I_{CC}	23	I_6	12 V	0 mA	1	$I_{CC} = I_6$	---	1.5	mA
	V_O 1/	24	V_{19}	12 V	0 mA	2	$V_O = V_{19}$	6.9472	7.0528	V
	V_{RLINE} 2/	25 26	V_{20} V_{21}	8.5 V 40 V	0 mA 0 mA	2 2	$V_{RLINE} = V_{19} - V_{20}$ $V_{RLINE} = V_{19} - V_{21}$	-0.2 -0.78	0.2 0.78	mV mV
	V_{RLOAD} 2/	27 28	V_{22} V_{23}	12 V 12 V	0 mA -10 mA	2 2	$V_{RLOAD} = V_{22} - V_{23}$	-2.8	2.8	mV
	I_{MIN}	29 30	V_{24} I_7	0 mA 0 mA	2 mA V_{25}	2 2	$V_{25} = V_{24} - 50$ mV $I_{MIN} = I_7$	--- -1.2	--- ---	mA
	V_{SHUNT} 2/ 3/	31 32	V_{26} V_{27}	0 mA 0 mA	1.2 mA 10 mA	2 2	$V_{SHUNT} = V_{27} - V_{26}$	-9.24	9.24	mV
	$\frac{\Delta V_O}{\Delta T}$	33					$\frac{\Delta V_O}{\Delta T} = \frac{V_{19} - V_1}{80}$	---	5	ppm/°C
4 T_A = +25°C	N_O 4/	34	N_1	15 V	0 mA	---	See figure 7	---	4	µV RMS

1/ Reference voltage is measured immediately after device is turned ON. Changes due to chip warm up are typically less than 0.005 percent.

2/ Line, load, and shunt mode regulation are measured on a pulse basis. Pulse is not to exceed 30 ms.

3/ Shunt voltage is measured with the input open. With the input connected, the shunt mode current can be reduced to 0 mA. Load regulation will remain the same.

4/ RMS noise is measured with a two pole high pass filter at 10 Hz and a two pole low pass filter at 1 kHz.

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TABLE III. Group A inspection for device type 09.

Subgroups	Symbol	Test no.	Measurement value	Adapter pin number See figure 6			Pin measurement	Equations	Limits		Unit
				1	2	3			Min	Max	
1	I_{CC}	1	I_1	15 V	0 mA	0 mA	1	$I_{CC} = I_1$	---	1.7	mA
T_A = +25°C	V_O 1/	2	V_1	15 V	0 mA	0 mA	2	$V_O = V_1$	9.950	10.050	V
	V_{RLINE} 2/	3	V_2	11.5 V	0 mA	0 mA	2				
		4	V_3	14.5 V	0 mA	0 mA	2	$V_{RLINE} = V_3 - V_2$	-0.12	0.12	mV
		5	V_4	40 V	0 mA	0 mA	2	$V_{RLINE} = V_3 - V_4$	-0.5	0.5	mV
	V_{RLOAD} 2/	6	V_5	15 V	0 mA	0 mA	2				
		7	V_6	15 V	-10 mA	0 mA	2	$V_{RLOAD} = V_5 - V_6$	-2.5	2.5	mV
	I_{MIN}	8	V_7	0 mA	2 mA	0 mA	2	$V_8 = V_7 - 50$ mV	---	---	mA
		9	I_2	0 mA	V_8	0 mA	2	$I_{MIN} = I_2$	-1.5	---	
	V_{SHUNT} 2/ 3/	10	V_9	0 mA	1.7 mA	0 mA	2				
		11	V_{10}	0 mA	10 mA	0 mA	2	$V_{SHUNT} = V_9 - V_{10}$	-8.3	8.3	mV
	V_{TRIM}	12	V_{11}	15 V	0 mA	0 mA	3	$V_{TRIM} = V_{11}$	4.3	---	V
	I_{OS}	13	I_3	15 V	0 V	0 mA	2	$I_{OS} = I_3$	-50	---	mA

See footnotes at end of table.

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TABLE III. Group A inspection for device type 09 - Continued.

Subgroups	Symbol	Test no.	Measurement value	Adapter pin number See figure 6			Pin measurement	Equations	Limits		Unit
				1	2	3			Min	Max	
2	I_{CC}	14	I_4	15 V	0 mA	0 mA	1	$I_{CC} = I_4$	---	1.7	mA
T_A = +125°C	V_O 1/	15	V_{12}	15 V	0 mA	0 mA	2	$V_O = V_{12}$	9.945	10.055	V
	V_{RLINE} 2/	16	V_{13}	11.5 V	0 mA	0 mA	2				
		17	V_{14}	14.5 V	0 mA	0 mA	2	$V_{RLINE} = V_{14} - V_{13}$	-0.18	0.18	mV
		18	V_{15}	40 V	0 mA	0 mA	2	$V_{RLINE} = V_{14} - V_{15}$	-1.0	1.0	mV
	V_{RLOAD} 2/	19	V_{16}	15 V	0 mA	0 mA	2				
		20	V_{17}	15 V	-10 mA	0 mA	2	$V_{RLOAD} = V_{17} - V_{16}$	-4.0	4.0	mV
	I_{MIN}	21	V_{18}	0 mA	2 mA	0 mA	2	$V_{19} = V_{18}$ - 50 mV	---	---	mA
		22	I_5	0 mA	V_{19}	0 mA	2	$I_{MIN} = I_5$	-1.7	---	
	V_{SHUNT} 2/ 3/	23	V_{20}	0 mA	1.7 mA	0 mA	2				
		24	V_{21}	0 mA	10 mA	0 mA	2	$V_{SHUNT} = V_{21} - V_{20}$	-12.4	12.4	mV
	$\frac{\Delta V_O}{\Delta T}$	25						$\frac{\Delta V_O}{\Delta T} = \frac{V_{10} - V_1}{100}$	---	5	ppm/°C

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TABLE III. Group A inspection for device type 09 - Continued.

Subgroups	Symbol	Test no.	Measurement value	Adapter pin number See figure 6			Pin measurement	Equations	Limits		Unit
				1	2	3			Min	Max	
3	I_{CC}	26	I_6	15 V	0 mA	0 mA	1	$I_{CC} = I_6$	---	2.0	mA
$T_A = -55°C$	V_O 1/	27	V_{22}	15 V	0 mA	0 mA	2	$V_O = V_{22}$	9.946	10.054	V
	V_{RLINE} 2/	28	V_{23}	11.5 V	0 mA	0 mA	2				
		29	V_{24}	14.5 V	0 mA	0 mA	2	$V_{RLINE} = V_{24} - V_{23}$	-0.18	0.18	mV
		30	V_{25}	40 V	0 mA	0 mA	2	$V_{RLINE} = V_{24} - V_{25}$	-1.0	1.0	mV
	V_{RLOAD} 2/	31	V_{26}	15 V	0 mA	0 mA	2				
		32	V_{27}	15 V	-10 mA	0 mA	2	$V_{RLOAD} = V_{27} - V_{26}$	-4.0	4.0	mV
	I_{MIN}	33	V_{28}	0 mA	2 mA	0 mA	2	$V_{29} = V_{28} - 50$ mV	---	---	mA
		34	I_7	0 mA	V_{29}	0 mA	2	$I_{MIN} = I_7$	-1.7	---	
	V_{SHUNT} 2/ 3/	35	V_{30}	0 mA	1.7 mA	0 mA	2				
		36	V_{31}	0 mA	10 mA	0 mA	2	$V_{SHUNT} = V_{31} - V_{30}$	-12.4	12.4	mV
	$\frac{\Delta V_O}{\Delta T}$	37						$\frac{\Delta V_O}{\Delta T} = \frac{V_{27} - V_1}{80}$	---	5	ppm/°C
4	N_O 4/	38	N_1	15 V	0 mA	0 mA	---	See figure 7	---	6	µV RMS
$T_A = +25°C$											

1/ Reference voltage is measured immediately after device is turned ON. Changes due to chip warm up are typically less than 0.005 percent.

2/ Line, load, and shunt mode regulation are measured on a pulse basis. Pulse is not to exceed 30 ms.

3/ Shunt voltage is measured with the input open. With the input connected, the shunt mode current can be reduced to 0 mA. Load regulation will remain the same.

4/ RMS noise is measured with a two pole high pass filter at 10 Hz and a two pole low pass filter at 1 kHz.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein. When the accelerated test conditions (T_A = +200°C or +250°C) are used for any test on a given inspection lot, they shall be used for all pertinent product assurance tests (screening (4.2), and group C inspection (4.4.3)) made on the lot.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Burn-in (method 1015 of MIL-STD-883), for class B and class S devices, test condition C, using the circuit shown on figure 9.

NOTE: If accelerated high-temperature test conditions are used, the device manufacturer shall ensure that at least 85 percent of the applied voltage is dropped across the device at temperature. The device is not considered functional under accelerated test conditions.

b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

c. Percent defective allowable (PDA) for class S and class B devices shall be as specified in MIL-M-38510, based on failures from group A, subgroup 1 test after cooldown as final electrical test in accordance with method 5004 of MIL-STD-883, and with no intervening electrical measurements. If interim electrical parameter tests are performed prior to burn-in, failures resulting from pre burn-in screening may be excluded from the PDA. If interim electrical parameter tests prior to burn-in are omitted, then all screening failures shall be included in the PDA. The verified failures of group A, subgroup 1 after burn-in divided by the total number of devices submitted for burn-in in that lot shall be used to determine the percent defective for that lot, and the lot shall be accepted or rejected based on the PDA for the applicable device class.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Electrical test requirements shall be as specified in table II herein.

b. Subgroups 6, 7, 8, 9, 10, and 11 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

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b. Steady-state life test for class S devices shall be in accordance with table IIa (subgroup 5) of method 5005 of MIL-STD-883, using the circuit shown on figure 9.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table IV herein.

b. Steady-state life test for class B devices (method 1005 of MIL-STD-883), test condition C, using the circuit shown on figure 9.

TABLE IV. Group C end-point electrical parameters, T_A = +25°C.

Device type	Characteristics	Symbol	Delta limits 1/	Limits		Unit
				Min	Max	
01, 04,05	Reference voltage V_S = 30 V, T_A = +25°C	V_{R1}, V_{R2}, V_{R3}	±2 mV	6.80	7.10	V
02,06	Reference voltage T_A = +25°C	V_{R1}, V_{R2}, V_{R3}	±3 mV	6.70	7.20	V
03	Reference voltage V_S = 15 V, T_A = +25°C	V_O	±3 mV	9.970	10.030	V
07	Reference voltage T_A = +25°C V_{IN} = 10 V	V_O	±3 mV	4.95	5.05	V
08	Reference voltage T_A = +25°C V_{IN} = 12 V	V_O	±3 mV	6.95	7.05	V
09	Reference voltage T_A = +25°C V_{IN} = 15 V	V_O	±3 mV	9.95	10.05	V

1/ Delta limits apply to the measured value (see delta limit definition in MIL-M-38510).

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4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883.

a. End-point electrical parameters shall be as specified in table II herein.

b. A special subgroup shall be added to the group D inspection requirements and shall consist of the tests, conditions, and limits specified for subgroup 5 of group A. The accept sample number for the subgroup shall be 15 devices with 0 failures for both classes.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables. Electrical test circuits as prescribed herein or in the referenced test methods of MIL-STD-883 shall be acceptable. Other test circuits shall require the approval of the qualifying activity.

4.5.1 Voltage and current. All voltage values given are referenced to the ground terminal of the device under test (DUT). Current values given are for conventional current and are positive when flowing into the referenced terminal.

4.5.2 Life test and burn-in cooldown procedure. When devices are measured at +25°C following application of the steady-state life or burn-in test condition, they shall be cooled to within +10°C of their power stable condition prior to removal of the bias.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logisitic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

c. Complete or Identifying Part Number (PIN) (see 6.6).

d. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

e. Requirements for certificate of compliance, if applicable.

f. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable.

g. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action and reporting of results, if applicable.

h. Requirements for product assurance options.

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i. Requirements for special lead lengths, or lead forming, if applicable. These requirements shall not affect the part number.

j. Requirements for "JAN" marking.

6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510 and MIL-STD-1331.

6.4 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.5 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	LM199A
02	LM129A
03	REF-10
04	LM199
05	LM199A-20
06	LM129B
07	LT1021-5
08	LT1021-7
09	LT1021-10

6.6 Part or Identifying Number (PIN). The PIN is created as specified in MIL-M-38510.

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI
Navy - OS, SH
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Preparing activity:
Air Force - 17

Agent:
DLA - ES

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-38510/124C	MICROCIRCUITS, LINEAR, PRECISION VOLTAGE REFERENCES, MONOLITHIC SILICON

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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